CSA06507-Generative AI

Lab Experiment's

.

Unit 2: Lab Programs for Prompting

Experiment 10: SQL Query Generation Using Structured Prompting

Aim:

To design structured prompts that provide database schema, table details, and requirements to a Large Language Model (LLM) to generate appropriate SQL queries and execute them to retrieve the required information.

Algorithm

1. Start the program.
2. Create a database using SQLite.
3. Create a table named Students with student details.
4. Insert sample student records into the table.
5. Design a structured prompt containing the database schema and query requirements.
6. Generate SQL queries based on the given requirements.
7. Execute the SQL queries using Python.
8. Retrieve and display the results.
9. Validate the generated SQL queries.
10. Stop the program.

Python Program

Python

Run

```
import sqlite3


# Step 1: Create database
conn = sqlite3.connect(":memory:")
cursor = conn.cursor()


# Step 2: Create Students table
cursor.execute("""
CREATE TABLE Students (
```

```
    ID INTEGER,
    Name TEXT,
    Department TEXT,
    Marks INTEGER
)
""")

# Step 3: Insert records
students = [
    (1, 'Ravi', 'ECE', 85),
    (2, 'Priya', 'CSE', 92),
    (3, 'Kiran', 'ECE', 78),
    (4, 'Anu', 'CSE', 88),
    (5, 'Rahul', 'EEE', 75)
]

cursor.executemany(
    "INSERT INTO Students VALUES (?, ?, ?, ?)",
    students
)

# Step 4: Structured Prompt
prompt = """
Database: Student Database

Table: Students
```

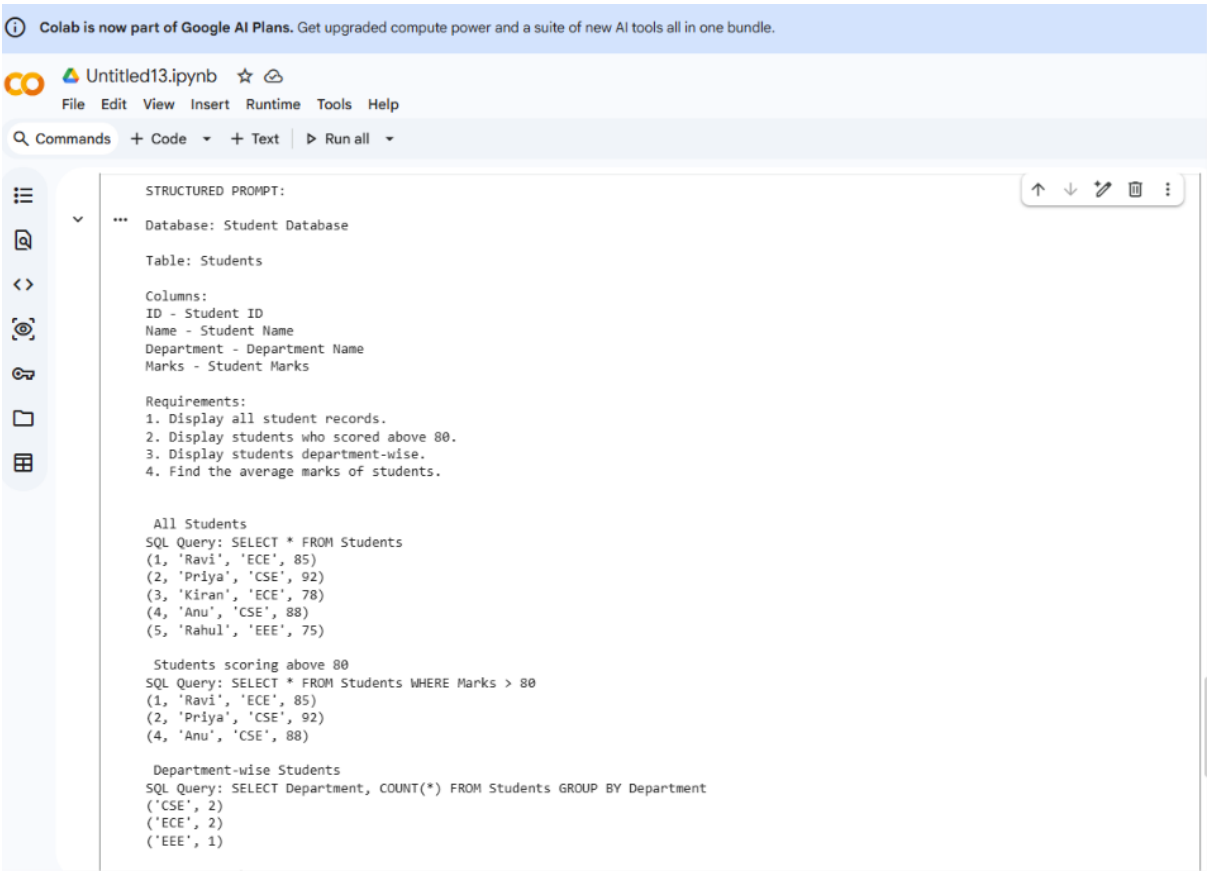
Input

The sample database contains the following student records:

ID	Name	Department	Marks
1	Ravi	ECE	85

2	Priya	CSE	92
3	Kiran	ECE	78
4	Anu	CSE	88
5	Rahul	EEE	75

Output



Result

Thus, SQL query generation using structured prompting was successfully implemented in Google Colab using Python and SQLite. A structured prompt was designed with database schema, table details, and query requirements. Appropriate SQL queries were generated, executed, and validated successfully with the expected results.

ChatGPT can make mistakes. Check important info.